Data Analytics
Exercise 1: SQL fundamentals

1. Write a SQL query to retrieve all columns from the employees table

```
SELECT *
From employees_db;
```

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. Write a SQL query to find all the unique departments in the employees table.

```
SELECT Distinct department
From employees_db;
```

department
IT
HR
Finance
Marketing

3. Write a SQL query to retrieve all employee's first and last names, order by salary in descending order.

```
SELECT first_name,
       last_name,
       Salary
From   employees_db
order by Salary desc;
```

first-name	last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Daniel	Clark	53000
Sarah	Brown	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4. Write a SQL query to retrieve the top 5 highest-paid employees

```
SELECT first_name,
       last_name,
       Salary
From   employees_db
order by Salary desc
Limit 5;
```

first_name	last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Daniel	Clark	53000

5. Write a SQL query to find employees who work in the IT department

```
SELECT first_name,
       last_name,
       department
FROM   employees_db
WHERE  department = 'IT';
```

first_name	last_name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6. Write a SQL query to find employees who work in the Finance depar
AND have a salary greater than 58,000.

```
SELECT first_name,
       last_name,
       department,
       Salary
From   employees_db
WHERE  department = 'Finance' AND Salary > 58000;
```

first_name	last_name	department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000

7. Write a SQL query to find employees who work in the HR department OR the Marketing department

```
SELECT first_name,
       last_name,
       department
From   employees_db
WHERE  department = 'HR' OR department = 'Marketing;
```

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

8. Write a SQL query to find employees who do not work in the IT department

```
SELECT first_name,
       last_name,
       department
From   employees_db
WHERE  department ~~NOT~~ <> 'IT';
```

first-name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. Write a SQL query to find employees who are in the HR, IT or Finance departments

```
SELECT first_name,
       last_name,
       department
FROM   employees_db
WHERE  department IN (HR, IT, Finance);
```

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

10. Write a SQL query to find employees who are in the IT department, have a salary greater than 50000 and are located in New York

```
SELECT first_name,
       last_name,
       department,
       Salary,
       City
From   employees_db
WHERE department = 'IT' AND Salary > 50000 AND City = New York;
```

First_name	last_name	department	salary
John	Doe	IT	55000
Sarah	~~Smith~~ Brown	IT	53000

11. Write a SQL query to retrieve the first and last names of employees who work in the Finance or Marketing department, earn more than 52000, and order the results by Salary in descending order.

```
SELECT first_name,
       last_name,
       department,
       Salary
From   employees_db
WHERE  department = 'Finance' or department = 'Marketing' AND
       Salary > 52000
ORDER BY Salary desc;
```

first_name	last_name	department	Salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

12. Write a SQL query to find all the unique cities where employees work, excluding those in the IT and HR departments.

```
SELECT distinct city,
       department
FROM   employees_db
WHERE department IN (Finance, Marketing);
```

City

department	City
Finance	Los Angeles
Marketing	San Francisco
Finance	Houston
Marketing	Chicago

13. Write a SQL query to retrieve employees who are not in the Finance department, have a salary greater than 50 000 and order the results by hire date in ascending order.

```
SELECT first_name,
       last_name,
       department,
       Salary,
       hire_date
from   employees_db
WHERE department <> 'Finance' AND salary > 50000
Order by hire_date asc;
```

first_name	last_name	department	Salary	hire_date
Emily	Davis	IT	62000	2015-02-14
David	White	Marketing	52000	2016-04-10
Jessica	Moore	HR	51000	2018-05-22
John	Doe	IT	55000	2018-06-15
Sarah	Brown	IT	53000	2021-03-25
Daniel	Clark	Marketing	53000	2022-06-01

14. Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department

```
SELECT first_name,
       last_name,
       City,
       department
FROM   employees_db
WHERE City IN ('Chicago','Los Angeles') AND department IN ('IT','Marketing')
Limit 3;
```

first_name	last_name	City	department
John	Doe	New York	IT
Jane	Smith	Chicago	HR
Mike	Johnson	Los Angeles	Finance